

Beefy Concentrated Liquidity Strategy Audit Summary

Prepared by Cyfrin

Version 2.0

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Contents

1 About Cyfrin 2

2 About This Document 2

3 Disclaimer 2

4 Copyright and License 2

5 Audit Overview 3

6 Audit Scope 3

7 Risk Classification 4

8 Protocol Summary 4

9 Attestation 4

1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Beefy Finance on Beefy Concentrated Liquidity Strategy. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated April 6, 2024, was delivered to Beefy Finance, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 11 days, the Cyfrin team conducted an audit on the Beefy Concentrated Liquidity Strategy code provided by Beefy Finance. The findings consist of 1 Critical, 1 High, 4 Medium & 10 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Beefy Concentrated Liquidity Strategy
Repository	experiments
Commit	14a313b76888...
Audit Timeline	Mar 18th - Apr 1st, 2024
Methods	Manual Review, Stateful Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	1	1	0
High Risk	1	0	1
Medium Risk	4	2	2
Low Risk	10	9	1
Informational	9	6	3
Gas Optimizations	9	4	5
Total	34	22	12

6 Audit Scope

The following contracts were included in the scope for this audit:

```
contracts/protocol/beefy/StratFeeManagerInitializable.sol
contracts/protocol/concliq/StrategyProtocol.sol
contracts/protocol/concliq/StrategyPassiveManagerUniswap.sol
contracts/protocol/vault/BeefyVaultConcLiq.sol
contracts/protocol/vault/BeefyVaultConcLiqFactory.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Beefy is a Decentralized, Multichain Yield Optimizer that allows its users to earn compound interest on their crypto holdings. To expand its functionality, Beefy developed a strategy to optimize yields obtained through Uniswap V3 LPs.

Uniswap V3 introduced the concept of concentrated liquidity, where users can decide the price range in which they want to provide liquidity, improving capital efficiency. This came with two main limitations:

- LP fees do not auto-compound, forcing users to manually claim their earned fees and then compound them, leading to additional expenses in gas.
- The price range is static: users do not have the option to provide liquidity in a dynamic price range, such as one that includes the current price.

Beefy's concentrated liquidity strategy intends to address these two limitations. The strategy provides liquidity within a set width around the current price, and this main position is updated each time a deposit or withdrawal is performed, at which point fees are compounded. As fees are always paid in one token, the excess token is provided in a narrower alternate position next to the current price.

The strategy is implemented in StrategyPassiveManagerUniswap; each strategy instance is deployed through the StrategyFactory contract using a Beacon proxy pattern. The strategy is operated by a Vault: every deposit and withdrawal is done through the Vault, compounded fees are locked for a given period before being available for withdrawal, and part of the earned fees are sent to a recipient specified when harvesting (by default `tx.origin`), Beefy and the strategist.

9 Attestation

Cyfrin attests that it performed a security review of the Beefy Concentrated Liquidity Strategy code provided by Beefy Finance.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-04-06-cyfrin-beefy-finance-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Beefy Finance has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=1693d8b053624d9dd6938eeb6fe51b2a086bfe609861cc1cbe4ccaa28cb9e90f
F=2024-04-06-cyfrin-beefy-finance-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.